Kushagra Pandya

AI ENGINEER

Google Scholar

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PROFESSIONAL SUMMARY

AI Engineer with 2.5+ years building and deploying production AI systems: computer vision, model serving, document classification and extraction, and liveness and deepfake detection. Now building agentic AI pipelines (LangGraph, CrewAI).

WORK EXPERIENCE

TSS Consultancy Pvt Ltd

03/2026 - Present

Software Engineer - AI

Rajkot, India

- Built a passive liveness-detection model against poster, 2D-mask, cropped-face and replay attacks: **89% average accuracy** on held-out benchmark datasets, ~0.5s inference, ~1,000 requests/day while in client use (07/2025 to 05/2026).
- Raised photo-match accuracy by **2 points (threshold 65) and 4 points (threshold 70)** and document-extraction accuracy by **~2% (relative)** with pre- and post-processing fixes, no retraining; fine-tuned the Aadhaar-number masking model on new data.

TSS Consultancy Pvt Ltd

01/2024 - 02/2026

Associate Software Engineer - AI

Rajkot, India

- Built and deployed identity-document classification and detection (YOLO, ONNX, custom inference pipeline): **98.92% accuracy** on a ~3K-image in-house benchmark, serving **~50K requests/day** on average at ~0.5s model inference and ~1.5s end-to-end latency.
- Built a deepfake detection model: **92%** on open-dataset test data, ~1s inference, ONNX, Dockerized API (deployed, not in client use).
- LLM/VLM fine-tuning PoCs (address parsing; licence extraction ~90%); not deployed, because of size and latency.

PROJECTS

football-detect-serve

Four YOLO26 detectors served as speed/accuracy presets (ONNX Runtime, FastAPI, Docker): test mAP50-95 0.400 to 0.477 at 16 to 177 ms per frame on CPU. ONNX was 1.5x faster than torch; INT8 cost 0.03 to 0.06 mAP50-95 with no CPU speedup.

football-team-tracker

Tracks players in broadcast video (YOLO26, Ultralytics trackers) and assigns teams by shirt colour (99.0% correct). Seven trackers benchmarked: best 63.6 HOTA on my detector vs 84.2 with perfect boxes.

SmartIngest

Multi-agent pipeline (LangGraph, FastAPI, Gemini) that classifies, extracts, validates and routes documents, with a deterministic router, injection and PII guardrails, and an eval-as-CI-gate harness (72 tests).

EDUCATION

B.Tech, Computer Engineering (Specialization in Artificial Intelligence) Marwadi University — 2020 - 2024

Rajkot, India

SKILLS

AI/ML & LLMs: Computer Vision, OCR, Fine-Tuning (LoRA/QLoRA), Model Optimization, Agentic AI, LLMs, RAG, Evals

Languages & Frameworks: Python, SQL, PyTorch, Ultralytics (YOLO), OpenCV, Hugging Face, scikit-learn, LangChain, LangGraph, CrewAI, LangSmith, Pydantic, Flask, FastAPI, Streamlit, PaddleOCR, Tesseract

LLM APIs: Azure OpenAI API, Gemini, Groq, LiteLLM **Vector Databases:** ChromaDB, Pinecone, Pgvector

MLOps & Deployment: Docker, ONNX Runtime, MLflow, Weights & Biases, Locust, GitHub Actions, Hugging Face Spaces, Git/GitHub